STIMULATION CONTROL · FIELD GUIDE

Stimulation Control Quick Start

Connect to the device, run stimulation by hand, run a saved protocol, and build your own. Same app, same steps, on Windows and macOS.

Before you start

- The device (miniBCS / Teensy) plugged into the computer by USB and powered on
- Stimulation Control installed — `Stimulation Control.exe` on Windows, `Stimulation Control.app` on macOS
- Optional: a protocol file (`.xlsx`) in the `Protocols` folder next to the app, e.g. `protocolC.xlsx`

1 Connect to the USB port

Do this first — everything else needs an active connection.

1. Open Stimulation Control. In the **Connection** box at the top, click `Refresh Ports` if your device isn't already listed.
2. Choose your device from the **Port** dropdown.
3. Click `Connect`. The label changes to `Connected: <port>` and the button becomes `Disconnect`.

WINDOWS PORT NAME	MACOS PORT NAME
looks like `COM3`, `COM4`	looks like `/dev/cu.usbmodem14201`

> Not sure which port is the device? Unplug it, click `Refresh Ports`, and see which entry disappears. Click `Disconnect` any time to let go of the port.

2 Run manual stimulation

Requires an active connection. Identical on Windows and macOS.

1. In the **Manual Control** box, set **Frequency** (0–500 Hz).
2. Check the **Stimulus** box to turn stimulation on at that frequency.
3. Uncheck it to turn stimulation off.

Status reads: ● ON ● OFF

> **0 Hz is not off** — it means the output stays continuously on instead of pulsing. Uncheck **Stimulus** to actually stop it.

> Every time you check **Stimulus**, a new recording starts automatically in the `Data` folder, named by time and frequency (e.g. `20261019T140501_20Hz.csv`), and stops when you uncheck it.

3 Run a saved protocol

Requires an active connection. Identical on Windows and macOS.

1. Click `Open File` and choose a protocol spreadsheet (e.g. from the `Protocols` folder). Its steps fill the table.
2. Click `Download Protocol` to send it to the device. Each command sent appears in the **Debug** panel.
3. Click `Run Protocol` to start. The button becomes `Abort Protocol` while it runs.
4. Click `Abort Protocol` to stop early, or let it finish on its own.

> A recording starts automatically the moment you click **Run Protocol**, named by time and protocol (e.g. `20261019T140930_protocolC.csv`), and stops when the run ends or is aborted.

4 Edit a protocol & save it as a new one

Identical on Windows and macOS.

Column	Meaning
`Duration (s)`	How long this step lasts, in seconds
`Pulse`	`On` or `Off` — whether stimulation runs during this step
`Width (ms)`	Pulse width in milliseconds (used only when Pulse is On and Freq is above 0)
`Freq (Hz)`	Pulse frequency — **0 Hz means continuously on**, not off

1. With a protocol loaded, double-click a cell to edit it (pick `On`/`Off` from the dropdown for **Pulse**).
2. Right-click the table for **Add Row Below**, **Duplicate Selected Row(s)**, or **Delete Selected Row(s)** — or use `Insert`, `Ctrl` + `D`, `Delete`.
3. Click `Save Protocol` and type a new file name, e.g. `protocolD.xlsx`. The original file is left untouched.
4. To test it on the device, click `Download Protocol` again — editing the table doesn't resend it by itself.

Windows vs macOS, at a glance

	Windows	macOS
Opening the app	Double-click `Stimulation Control.exe`	Double-click `Stimulation Control.app`. First time only: if macOS says it's from an unidentified developer, **right-click → Open** instead, then confirm **Open**.
Port names	`COM3`, `COM4`, …	`/dev/cu.usbmodem14201`, …
`Data` & `Protocols` **folders**	Next to `Stimulation Control.exe`	Next to `Stimulation Control.app` in Finder (not inside it)
Everything else	Connecting, manual stimulation, running and editing protocols work exactly the same way on both.	